Shreyas Patil

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EDUCATION

B.Tech, Computer Science & Engineering, *IIIT Pune* *Aug 2023 – May 2027*
CGPA: 7.27 / 10.0

Senior Secondary (Science), *Sudhakar Naik Jr. College, Akola* *2021 – 2023*
Class XII, Maharashtra State Board | Percentage: 78.50%

EXPERIENCE

Software Developer & iOS Operations Lead, *Shreepad Seva Mandal (Nonprofit App)* *Apr 2026 – Aug 2026*
Pune, Maharashtra (Remote)

- Spearheaded **iOS release engineering** across **98 TestFlight builds**, scaling the app to **1,000+ live users**.
- Resolved production **iOS APNs push notification** delivery bugs and **Firebase Auth** race conditions.
- Synchronized **iOS/Android** parity via **Capacitor** native bridges, shipping fixes with **20+ beta testers**.

Software Development Engineer Intern, *Xelron* *Mar 2026 – Apr 2026*
Bengaluru, Karnataka (Remote)

- **Project Gauss:** Designed **HLE-style** adversarial evaluation tasks and validation gates across **9 domains**.
- **Project Einstein:** Validated **STEM training data** against multi-stage rubric, schema, and **LaTeX** checks.
- Benchmarked **GPT and Claude** on **42+ accepted prompts**, isolating multi-step reasoning failure modes.

PROJECTS

RecoveryOS – Autonomous Payment Recovery Engine | *Python, FastAPI, SQLite, React 19, Gemini API* [GitHub]

- Engineered idempotent **Razorpay webhooks** (**HMAC-SHA256**) with **SQLite WAL** claims to prevent duplicate billing.
- Coupled fail-closed **Gemini 3.5 Flash** diagnosis with a **6-rule policy gate** and **RBI e-Mandate** EV optimization.
- Achieved **+INR 191.7K mean True EV lift** and **94.2% Oracle capture** via integer **Empirical-Bayes** shrinkage.
- Enforced zero-float financial safety across **1,753 Pytest tests** with **5 AST isolation suites** in **GitHub Actions CI**.

Temporal Memory Engine – Bitemporal AI Memory | *Python, FastAPI, PostgreSQL 16 (pgvector, RLS)* [GitHub]

- Architected a **bitemporal engine** in **PostgreSQL 16** with **RLS**, advisory locks, and coalesced **outbox queues**.
- Constructed a pure **S1–S10 conflict resolver** for out-of-order backfills, verified by **Hypothesis** property tests.
- Orchestrated **5-lane hybrid retrieval** (HNSW, BM25, 2-hop **graph CTE**, temporal) using **RRF** and decay in **Asyncio**.
- Delivered **96.7% LongMemEval accuracy** (**+62% update F1**), backed by **482 automated tests** and **strict Mypy**.

Skin Cancer Detection via Deep Transfer Learning | *Python, PyTorch, EfficientNet, ResNet, FastAPI* [GitHub] | [Live]

- Fused a **54.5M-param** 4-CNN binary ensemble via **6,144-d LayerNorm** and soft voting on **59K+ ISIC** images.
- Mitigated **7.4:1 class imbalance** via **Focal Loss** and spatial augmentations, achieving **0.960 ROC-AUC**.
- Calibrated the malignant threshold ($p > 0.28$), lifting melanoma recall **+15.8%** to **85.8%** (**93.8% precision**).
- Containerized a **FastAPI** & **React 19** app with threadpool inference, magic-byte checks, and **weights_only** guards.

TECHNICAL SKILLS

Languages	C++, Python, JavaScript, SQL
AI/ML	PyTorch, NumPy, Pandas, LLMs, LLM APIs, Transfer Learning, Computer Vision
GenAI	LangChain, LangGraph, RAG, Vectorless RAG, Guardrails, Evals
Backend	FastAPI, Asyncio, Node.js, Express.js, REST APIs, JWT, Firebase
Frontend & Mobile	React.js, Next.js, Capacitor, Streamlit, Tailwind CSS
Databases	MySQL, PostgreSQL, SQLite, MongoDB, Redis
Testing & DevOps	Pytest, Hypothesis, Mypy, Docker, Git, Linux, GitHub Actions CI/CD
Core CS	DSA, OOP, DBMS, LLD, Operating Systems, Computer Networks, SOLID Principles

ACHIEVEMENTS & CERTIFICATIONS

- **Cyber Cypher 5.0 Finalist (Team Lead)** – **Top 30 / 250+** for **ArcticOps** (NMIMS) *[Certificate] Mar 2026*
- **SQL & Data Engineering Foundations** – Certified by **takeUforward** in SQL & ETL *[Certificate] Oct 2026*